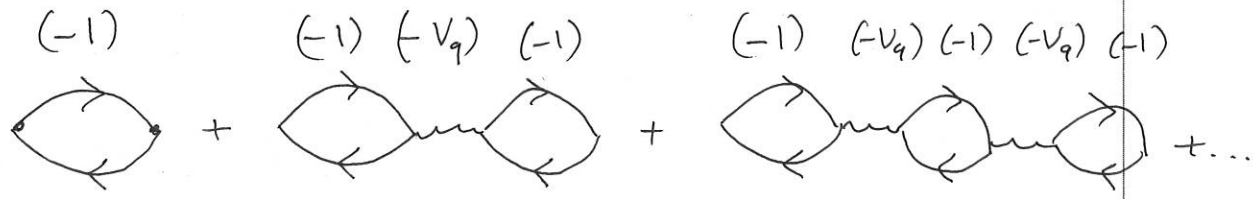
# Diagrams for charge and spin RPA

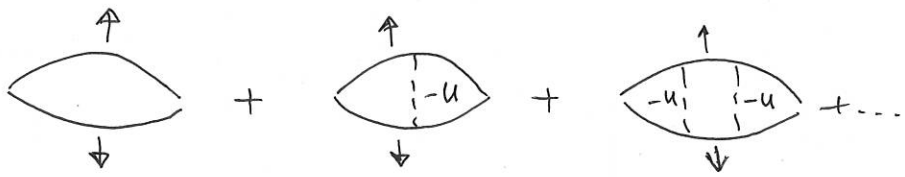



$$= -\tilde{\Pi} + (-\tilde{\Pi})(-V_q)(-\tilde{\Pi}) + (-\tilde{\Pi})^3(-V_q)^2 + \dots$$

$$= -\tilde{\Pi}\left(1 + \tilde{\Pi} V_q + \tilde{\Pi}^2 V_q^2 + \dots\right) = \frac{-\tilde{\Pi}}{1 - V_q \tilde{\Pi}}$$

The minus sign on the "bubble" element is because it forms closed fermion loop.

Let us imagine the situation where the loop is not present $\Rightarrow$ transverse spin susceptibility




There is no loop because the spin $\sigma = \uparrow, \downarrow$ is flipped from "up" to "down" orientation at the right-most vertex (and vice versa at the left-most)

$$\tilde{\Pi}_{\uparrow\downarrow} + \tilde{\Pi}_{\uparrow\downarrow}(-U)\tilde{\Pi}_{\uparrow\downarrow} + \tilde{\Pi}_{\uparrow\downarrow}(-U)\tilde{\Pi}_{\uparrow\downarrow}(-U)\tilde{\Pi}_{\uparrow\downarrow} + \dots$$

$$= \tilde{\Pi}_{\uparrow\downarrow}\left(1 - U\tilde{\Pi}_{\uparrow\downarrow} + U^2 \Pi_{\uparrow\downarrow}^2 + \dots\right)$$

$$= \frac{\tilde{\Pi}_{\uparrow\downarrow}}{1 + U\tilde{\Pi}_{\uparrow\downarrow}}$$

The absence of the fermion loop changes the sign in the denominator! (Charge and spin response ARE different)

Diagrams for density-density and spin-spin two-particle GFs

$$K_M(\vec{r},\tau) = -\frac{T_R \langle e^{-\beta \hat{H}_0} T_\tau \hat{\rho}(\vec{r},\tau)\, \hat{\rho}(0,0)\, S(\beta)\rangle}{T_R\left(e^{-\beta \hat{H}_0} S(\beta)\right)}$$

where $\hat{S}(\beta) = T_\tau \exp\left(-\int_0^\beta d\tau_1 \hat{V}(\tau_1)\right) =$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_0^\beta d\tau_1 \ldots \int_0^\beta d\tau_n \left\{T_\tau\, \hat{V}(\tau_1) \ldots \hat{V}(\tau_n)\right\}$$

Here $\hat{H}_0$ is non-interacting Hamiltonian and $\hat{V}$ is the interaction.

In summing RPA series we previously ~~used~~ wrote

$$K_M = P + PVP + PVPVP + \ldots = P + PVK_M$$

$$K_M(q,\omega_n) = \frac{P_M(q,\omega_n)}{1 - V_q P_M(q,\omega_n)}$$

The RPA approximation consists in using $P_M \to P^{(1)}$ for the irreducible polarization propagator.

$$P^{(1)}(q,\omega) = \frac{2}{\beta V} \sum_{\vec{p},\, p_m} G_M^{(0)}(\vec{p}, ip_m)\, G_M^{(0)}(\vec{p}+\vec{q}, ip_m + i\omega_n)$$

Quick check of the static limit:

$$P^{(1)}(q\to 0, \omega_n = 0) = \frac{2}{\beta V}\sum_{\vec{p}, p_M} \frac{1}{ip_m - \xi_{\vec{p}}} \frac{1}{ip_m + \underset{\overset{\shortparallel}{0}}{i\omega_n} - \xi_{\vec{p}+\vec{q}}}$$

$$\xrightarrow[q\to 0]{} \frac{2}{\beta}\sum_{p_m}\int\frac{(d\vec{p})}{(2\pi)^3}\frac{1}{(ip_m - \xi_p)^2}$$

We know that $S_1(\tau) = \frac{1}{\beta}\sum_{\nu_\ell}\frac{e^{-i\nu_\ell\tau}}{i\nu_\ell - \varepsilon} = e^{-\tau\varepsilon} n_F(\varepsilon)$

$$\Rightarrow \frac{1}{\beta}\sum_{p_m}\frac{1}{(ip_m - \xi_p)^2} = \frac{\partial}{\partial\xi_p}\frac{1}{\beta}\sum_{p_m}\frac{1}{ip_m - \xi_p} =$$

$$= \frac{\partial}{\partial\xi_p}\lim_{\tau\to 0}\frac{1}{\beta}\sum_{p_m}\frac{e^{-ip_m\tau}}{ip_m - \xi_p} = \frac{\partial}{\partial\xi_p}\lim_{\tau\to 0}\left(e^{-\tau\xi_p} n_F(\xi_p)\right)$$

$$= \frac{\partial}{\partial\xi_p} n_F(\xi_p)$$

Thus $P^{(1)}(q\to 0, \omega_n = 0) = 2\int\frac{d\vec{p}}{(2\pi)^3}\frac{\partial}{\partial\xi_p} n_F(\xi_p) =$

$$= \frac{2\cdot 4\pi}{(2\pi)^3}\int dp\, p^2 \frac{\partial}{\partial\xi_p} n_F(\xi_p) = \left(\text{switch to integration over } \xi\right)$$

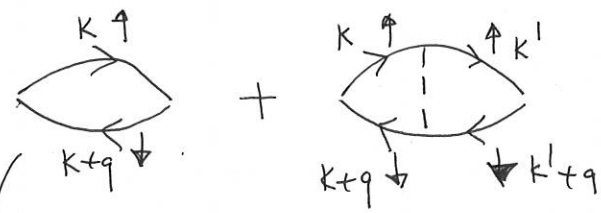
$$= \frac{8\pi}{(2\pi)^3}\frac{p_F^2}{v_F}\underbrace{\int_{-\infty}^{\infty} d\xi \frac{\partial}{\partial\xi} n_F(\xi)}_{\substack{n_F(\infty) - n_F(-\infty) \\ \shortparallel \qquad\quad \shortparallel \\ 0 \qquad\quad 1}} = -\frac{m p_F}{\pi^2} = -2\nu.$$

Thus $P^{(1)}$ stands for polarization bubble <u>without</u> the Fermi-loop $(-1)$ sign.

Hence $K_\mu(q\to 0, \omega_n = 0) = \dfrac{P_\mu^{(1)}(q\to 0, \omega_n = 0)}{1 - V_q\cdot(-2\nu)} \to \dfrac{P^{(1)}(q, 0)}{1 + 2\nu V_q}$ screening!

Transverse

Spin susceptibility $\Pi_{\uparrow\downarrow}(\vec{q}, \omega_m)$

$\Pi_{\uparrow\downarrow} =$ 


$$\Pi^{(0)}_{\uparrow\downarrow}(\vec{q}, \omega_m) = \frac{1}{\beta V} \sum_{\vec{k}, k_n} G^{(0)}_{\uparrow}(\vec{k}, ik_n)\, G^{(0)}_{\downarrow}(\vec{k}+\vec{q}, ik_n + i\omega_m)$$

$$= \frac{1}{\beta} \sum_{k_n} \int \frac{d\vec{k}}{(2\pi)^3} \frac{1}{ik_n - \xi_{k\uparrow}} \frac{1}{ik_n + i\omega_m - \xi_{k+q\downarrow}}$$

$$= \frac{1}{\beta} \sum_{k_n} \int (d\vec{k}) \left( \frac{1}{ik_n - \xi_{k\uparrow}} - \frac{1}{ik_n + i\omega_m - \xi_{k+q\downarrow}} \right) \frac{1}{i\omega_m - \xi_{k+q\downarrow} + \xi_{k\uparrow}}$$

$= \left(\text{add } e^{-ik_n\tau} \text{ convergence factor and use result for } S_1(\tau)\right) =$

$$= \int (d\vec{k}) \frac{n_F(\xi_{k\uparrow}) - n_F(\xi_{k+q\downarrow})}{i\omega_m + \xi_{k\uparrow} - \xi_{k+q\downarrow}}$$

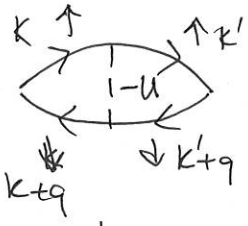
In the unpolarized system (no external magnetic field and no spontaneous magnetization) $\xi_{k\uparrow} = \xi_{k\downarrow}$.

Then $\Pi^{(0)}_{\uparrow\downarrow}(q \to 0, \omega_m = 0) = -\nu$ since in this case

$P^{(1)}(\vec{q}, \omega_n)$ and $\Pi^{(0)}_{\uparrow\downarrow}(\vec{q}, \omega_m)$ are given by the same convolution of single particle Green's functions $\frac{1}{\beta V}\sum_k G^{(0)}(\vec{k}, ik_n)\, G^{(0)}(\vec{k}+\vec{q}, ik_n + i\omega_n)$

(they do differ by spin degeneracy $\to$ factor of (2) in $P^{(1)}$)

$$\Pi^{(1)}_{\uparrow\downarrow}(\vec{q},\omega_n) =$$




The dashed line is interaction $(-U)$ which does not depend on momentum/frequency

But
$$\frac{1}{\beta V}\sum_{\vec{k}\,k_n}\frac{1}{\beta V}\sum_{\vec{k}'\,k_n'} G^{(0)}_{\uparrow}(\vec{k},ik_n)\,G^{(0)}_{\uparrow}(\vec{k}',ik_n')\,G^{(0)}_{\downarrow}(\vec{k}+\vec{q},ik_n+i\omega_m)\,G^{(0)}_{\downarrow}(\vec{k}'+\vec{q},ik_n'+i\omega_m)$$

$$= \left(\frac{1}{\beta V}\sum_{\vec{k},k_n} G^{(0)}_{\uparrow}(\vec{k},ik_n)\,G^{(0)}_{\downarrow}(\vec{k}+\vec{q},ik_n+i\omega_m)\right)^2$$

$$\Rightarrow \Pi^{(1)}_{\uparrow\downarrow} = -U\left(\Pi^{(0)}_{\uparrow\downarrow}\right)^2$$

So the spin RPA series goes as

$$\Pi_{\uparrow\downarrow} = \Pi^{(0)}_{\uparrow\downarrow} + (-U)\left(\Pi^{(0)}_{\uparrow\downarrow}\right)^2 + (-U)^2\left(\Pi^{(0)}_{\uparrow\downarrow}\right)^3 + \ldots =$$

$$= \Pi^{(0)}_{\uparrow\downarrow}\left\{1 - U\,\Pi^{(0)}_{\uparrow\downarrow} + U^2\left(\Pi^{(0)}_{\uparrow\downarrow}\right)^2 + \ldots\right\}$$

the series is sign-alternating

$$= \frac{\Pi^{(0)}_{\uparrow\downarrow}}{1+U\,\Pi^{(0)}_{\uparrow\downarrow}}$$

Quick check: $\omega_n = 0$, $q\to 0$ limit: $\Pi^{(0)}_{\uparrow\downarrow} \to -\nu$

$$\Pi_{\uparrow\downarrow} \to \frac{-\nu}{1-\nu U}$$

$\Rightarrow$ instability (Stoner) is possible once $1 = \nu U$ for $U>0$.